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Scenario – 30

SE LAB - 01

Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Passengers to submit a lost baggage claim by entering their PNR (Passenger Name Record) and Baggage Tag Number.	High	**Pass:** Claim is successfully created and a unique Claim ID is generated. **Fail:** Claim submission is allowed without a valid PNR or Tag Number.	Provides the core entry point for the passenger to report missing luggage.
FR-002	Functional	The system shall cross-reference passenger lost baggage reports	High	**Pass:** A match is identified between a claim and an unrouted scan log,	Automates the discovery and matching process to speed up

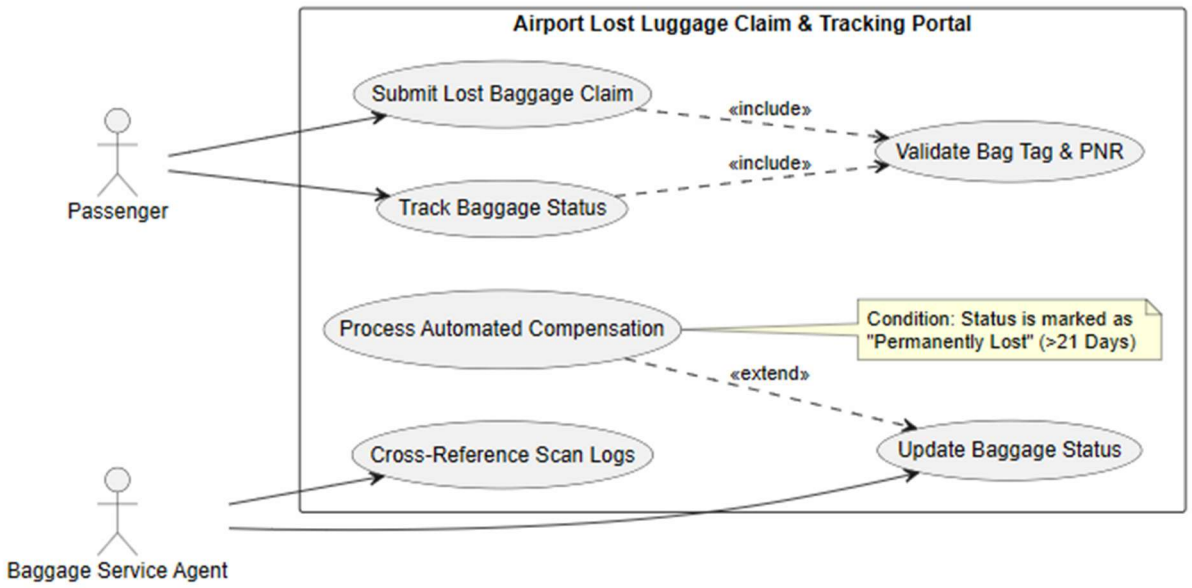
ID	Type	Description	Priority	Acceptance Criteria	Rationale
		against unrouted luggage scan logs across all connected airport terminals.		generating a recovery alert. **Fail:** The system fails to flag a matching tag number sitting in the unrouted database.	luggage recovery.
FR-003	Functional	The system shall allow Baggage Service Agents to manually update the tracking status and current physical location of a bag.	High	**Pass:** Agent updates status to "In Transit to Passenger," and the system logs the agent ID and timestamp. **Fail:** Status update saves without recording the	Ensures human oversight and accurate tracking during the physical recovery phase.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
				time or location data.	
FR-004	Functional	The system shall automatically send real-time SMS and email notifications to the Passenger whenever their luggage tracking status changes.	Medium	**Pass:** Notification is dispatched within 1 minute of a status change in the database. **Fail:** Status changes, but no notification is triggered.	Keeps the passenger informed, reducing anxiety and support desk call volume.
FR-005	Functional	The system shall initiate an automated compensation workflow if a baggage claim remains in the "Lost" state for more than 21 consecutive days.	Medium	**Pass:** A compensation form link is automatically emailed to the passenger on day 22. **Fail:** The claim remains open past 21 days	Ensures compliance with international aviation passenger rights and compensation laws.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
				without triggering the workflow.	
NFR-001	Performance	The baggage tracking timeline API must return complete multi-leg scan history in under 200 ms.	High	**Pass:** Load tests confirm 95th percentile latency is ≤ 200 ms under 1,000 concurrent requests. **Fail:** API response takes longer than 200 ms during peak load.	Ensures a seamless and responsive user experience when passengers are anxiously checking tracking updates.
NFR-002	Security	All Passenger Personally Identifiable Information (PII), including contact details and delivery addresses, must be	High	**Pass:** Database audits verify that all PII columns are fully encrypted and	Protects sensitive user data from breaches and complies with global data privacy regulations (e.g., GDPR).

ID	Type	Description	Priority	Acceptance Criteria	Rationale
		encrypted at rest using AES-256.		inaccessible in plaintext. **Fail:** Delivery addresses are stored in plaintext in the database.	

UML Diagram


Airport Lost Luggage Claim & Tracking Portal
Submit Lost Baggage Claim
«include»
Validate Bag Tag & PNR
«include»
Track Baggage Status
Process Automated Compensation
Condition: Status is marked as "Permanently Lost" (>21 Days)
«extend»
Cross-Reference Scan Logs
Update Baggage Status
Passenger
Baggage Service Agent

Use-Case Flow Specification

Use Case ID: UC-01

Use Case Name: Submit Lost Baggage Claim

Primary Actor: Passenger

Secondary Actor: Baggage Service Agent (Receives Alert)

- **Preconditions:**
    1. The Passenger has a valid flight booking (PNR) and a Baggage Tag Number provided at check-in.
    2. The Passenger has access to the airline's web portal or mobile application.
- **Postconditions:**
    1. A unique Claim ID is generated and linked to the Passenger's PNR.
    2. The bag's status is updated to "Reported Lost - Tracing" in the centralized database.
    3. The system initiates an automatic cross-reference search against unrouted luggage logs.
- **Main Success Scenario (MSS):**
    1. Passenger navigates to the "Lost Luggage" section of the airline portal.
    2. Passenger inputs their PNR, Last Name, and Baggage Tag Number.
    3. System triggers validation to ensure the PNR and Bag Tag match airline records (<<include>> Validate Bag Tag & PNR).
    4. System retrieves and displays the flight details and passenger information.
    5. Passenger fills in a description of the bag (color, brand, type) and provides a preferred delivery address and contact phone number.
    6. Passenger clicks "Submit Claim".
    7. System saves the data, generates a unique Claim ID, and sets the status to "Tracing".
    8. System sends an automated confirmation SMS and Email containing the Claim ID and a tracking link to the Passenger.

9. System alerts the Baggage Service Agent dashboard at the destination airport.

- **Alternate Flow (A1 - Invalid Credentials or Format):**
    - **3a.** The system detects that the PNR does not exist, or the Baggage Tag Number format is invalid/does not belong to the entered PNR.
    - **3b.** The system halts the submission process.
    - **3c.** The system displays an error message: "Invalid PNR or Baggage Tag Number. Please verify the details on your boarding pass and baggage receipt."
    - **3d.** The Passenger is prompted to re-enter the correct details or use a link to contact live support.